

SAT UZBEKISTAN

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1  Mark for Review



A certain hybrid car uses 39 gallons of gasoline to travel 1,950 miles. At this rate, how many kilometers does the car travel using 39 gallons of gasoline? (Use 1 mile = 1.6 kilometers.)

(A) 312

(B) 3,120

(C) 31,200

(D) 312,000



The function h is defined by $h(x) = 4x - 9$. What is the value of $h(-3)$?

(A) 21

(B) 13

(C) -13

(D) -21



$$10x = 110$$

$$8x - 86 = y$$

The solution to the given system of equations is (x, y) . What is the value of y ?

(A) 86

(B) 11

(C) 10

(D) 2



Which expression is equivalent to $7x^5 - 6x^4 + 9x^3$?

(A) $x^4(7x - 6)$

(B) $x^3(7x^2 - 6x + 9)$

(C) $9x^3(7x^2 - 6x + 1)$

(D) $6x^5(-6x^4 + 9x^3 + 1)$



$$y = 0$$

$$y = 8(x^2 - 36)$$

Which ordered pair (x, y) is a solution to the given system of equations?

(A) $(0, 6)$

(B) $(0, 36)$

(C) $(6, 0)$

(D) $(36, 0)$



A group of students purchase ice cream cones and ice cream sundaes. The cones cost \$1.40 each, and the cost for 19 sundaes is \$64.60. The total cost for the cones and sundaes is \$85.60. Which equation represents this situation, where x is the number of cones purchased and y is the number of sundaes purchased?

(A) $1.40x + 64.60y = 85.60$

(B) $64.60x + 1.40y = 85.60$

(C) $3.40x + 1.40y = 85.60$

(D) $1.40x + 3.40y = 85.60$

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67



7



Mark for Review

Hakim has a bag of 50 tiles, where 45 are labeled with a letter and 5 are blank. If Hakim selects a tile from the bag at random, what is the probability of selecting a blank tile? (Express your answer as a decimal or fraction, not as a percent.)

Answer Preview:

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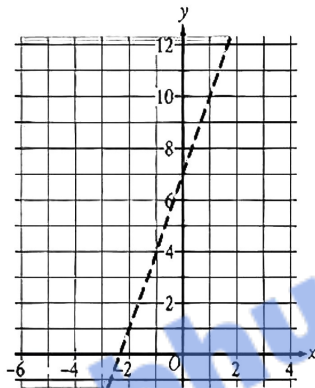
In $\triangle DEF$, the measure of $\angle D$ is 114° . Which of the following could be the measure, in degrees, of $\angle E$?

(A) 115

(B) 90

(C) 67

(D) 65



The shaded region shown represents the solutions to which inequality?

(A) $y > 3x + 7$

(B) $y > 3x - 7$

(C) $y < 3x + 7$

(D) $y > 3x - 7$



Which expression is equivalent to $20x^2 + 3 + 5x^2$?

(A) $25x^4 + 3$

(B) $25x^2 + 3$

(C) $28x^4$

(D) $28x^2$



A company opens an account with an initial balance of \$35,600.00 . The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function A , where $A(t)$ is the account balance, in dollars, t years after the account is opened. The account balance after 15 years is \$129,672.38 . Which equation could define A ?

(A) $A(t) = 35,600.00(1.09)^t$

(B) $A(t) = 35,600.00(0.09)^t$

(C) $A(t) = 94,072.38(1.09)^t$

(D) $A(t) = 94,072.38(0.09)^t$



What is the y-intercept of the graph of $y = 14^x$ in the xy -plane?

(A) $(0, 15)$

(B) $(0, 14)$

(C) $(0, 1)$

(D) $(0, -1)$

Examples

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
	.6667	.66
		0.67



13



Mark for Review

The function f is defined by $f(x) = 2x + b$, where b is a constant and $f(7) = 6$. What is the value of b ?

Answer Preview:

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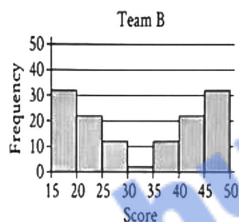
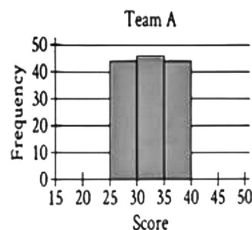
A number f is 120% greater than a positive number g . A number h is 90% less than the number f . The number h is how many times the number g ?

(A) 0.12

(B) 0.22

(C) 1.08

(D) 1.98

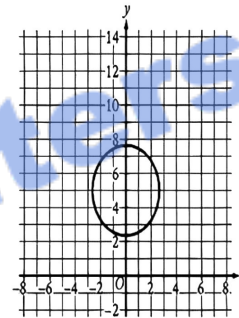


The histograms summarize the distributions of scores for team A and team B. Which statement best compares the standard deviations of scores for these teams?

- ☐ A The standard deviation of scores for team A is less than the standard deviation of scores for team B.
- ☐ B The standard deviation of scores for team A is greater than the standard deviation of scores for team B.
- ☐ C The standard deviation of scores for team A is equal to the standard deviation of scores for team B.
- ☐ D There is not enough information to compare the standard deviations.

3.5	3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	-1/3 -.3333 -0.333	-.33 -0.33

16 Mark for Review



Circle A shown is defined by the equation $x^2 + (y - 5)^2 = 7$. Circle B (not shown) has the same radius but is translated 91 units to the right. If the equation of circle B is $(x - h)^2 + (y - k)^2 = a$, where h , k , and a are constants, what is the value of $4a$?

Answer Preview:

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	$\frac{2}{3}$.6666 .6667 0.666 0.667	0.66 .66 0.67 .67



17



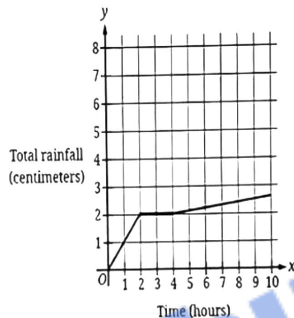
Mark for Review

$$h(t) = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is 40.96 feet above the ground when it is dropped at $t = 0$. How many seconds after being dropped does the function estimate the object will hit the ground?

Answer Preview:

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The graph shows the total amount of rainfall y , in centimeters, from the start of a 10-hour period, where x is the number of hours after the start of the period. Which of the following statements about the rainfall during this time period is true?

- (A) The rate of rainfall was 2 centimeters per hour between $x = 2$ and $x = 4$.
- (B) The rate of rainfall was the greatest between $x = 4$ and $x = 10$.
- (C) The rate of rainfall increased between $x = 0$ and $x = 2$.
- (D) The rate of rainfall was 0 centimeters per hour between $x = 2$ and $x = 4$.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .66 .6667 0.666	0.66 0.67

19



Mark for Review

For a set of three consecutive integers, the result of subtracting the greatest integer from the sum of the other two integers is 48. What is the value of the greatest of the three integers?

Answer Preview:

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A rectangular banner has an area of 2,300 square inches. A copy of the banner is made in which the length and width of the original banner are each increased by 20%. What is the area of the copy, in square inches?

(A) 2,320

(B) 2,340

(C) 2,760

(D) 3,312



$$(x - 42)^2 = 1$$

What is the sum of the solutions to the given equation?

(A) 42

(B) 43

(C) 84

(D) 85

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
	.6667	.66
	0.666	0.67
	0.667	.67



22



Mark for Review

$$5(kx - n) = -\frac{65}{14}x - \frac{85}{18}$$

In the given equation, k and n are constants and $n > 1$. The equation has no solution. What is the value of k ?

Answer Preview:

1



Mark for Review



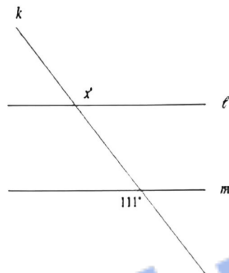
If $9 + x = 6$, what is the value of $-18 - 2x$?

(A) -12

(B) -2

(C) 6

(D) 24



Note: Figure not drawn to scale.

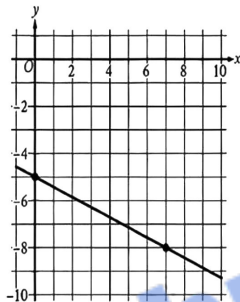
In the figure, line l is parallel to line m . What is the value of x ?

(A) 31

(B) 69

(C) 111

(D) 131



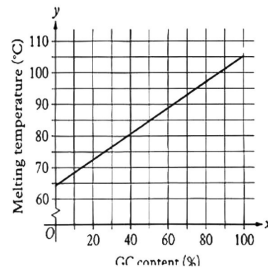
The graph of the linear function f is shown, where $y = f(x)$. Which equation defines f ?

(A) $f(x) = \frac{3}{7}x - 5$

(B) $f(x) = \frac{7}{3}x - 5$

(C) $f(x) = -\frac{3}{7}x - 5$

(D) $f(x) = -\frac{7}{3}x - 5$



The function t gives the estimated melting temperature, in degrees Celsius, of a DNA molecule as a function of the percent of guanine-cytosine (GC) content, x , in the DNA molecule. The graph of $y = t(x)$ is shown. According to the graph, which of the following statements is NOT true?

- (A) For each increase of x by 1, $t(x)$ increases by approximately $\frac{2}{5}$.
- (B) A DNA molecule with a GC content of 0% has an estimated melting temperature between 60°C and 65°C.
- (C) A DNA molecule with a GC content of 95% has an estimated melting temperature between 100°C and 105°C.
- (D) A DNA molecule with a GC content of 98% has an estimated melting temperature between 80°C and 90°C.



In the xy -plane, line s is parallel to line t . Line s is defined by $y + 16 = 7x$. What is the slope of line t ?

(A) -16

(B) -7

(C) 7

(D) 16



Scientists collected fallen acorns that each housed a colony of the ant species *P. ohioensis* and analyzed each colony's structure. For any of these colonies, if the colony has x worker ants, the equation $y = 0.67x + 2.6$, where $20 \leq x \leq 110$, gives the predicted number of larvae, y , in the colony. If one of these colonies has 39 worker ants, which of the following is closest to the predicted number of larvae in the colony?

(A) 29

(B) 42

(C) 54

(D) 128

$$y > 5x - 4$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

(A)

x	y
4	11
6	21
9	36

(B)

x	y
4	16
6	26
9	41

(C)

x	y
4	21
6	31
9	46

(D)

x	y
4	21
6	21
9	41

$$P = N(34 - C)$$

The given equation relates the variables P , N , and C . Which expression represents the value of C for distinct positive values of P and N ?

(A) $C + \frac{P}{N}$

(B) $34 + \frac{P}{N}$

(C) $34 - \frac{P}{C}$

(D) $34 - \frac{P}{N}$



$$x^2 - 15x + 12 = 0$$

What is the sum of the solutions to the given equation?

(A) 15

(B) 12

(C) 3

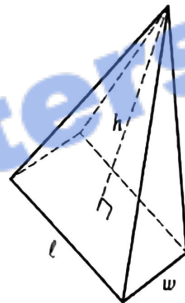
(D) 0

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
	.6667	.66
	0.666	0.67
	0.667	.67
1	-1/3	-.33



10

Mark for Review

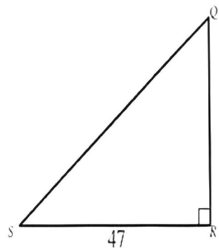


Note: Figure not drawn to scale.

The figure shown is a right rectangular pyramid, where $l = 18$ units, $w = 9$ units, and $h = 12$ units. What is the surface area, in square units, of the pyramid?

The function $f(x) = \frac{1}{7}(x - 6)^2 + 4$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- (A) The toy car's minimum height was 4 inches above the ground.
- (B) The toy car's minimum height was 6 inches above the ground.
- (C) The toy car's height was 4 inches above the ground when it started moving.
- (D) The toy car's height was 6 inches above the ground when it started moving.



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of QS ?

(A) $47\cos Q$

(B) $47\sin Q$

(C) $\frac{47}{\cos Q}$

(D) $\frac{47}{\sin Q}$

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5	
	3.50	3 1/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
	.66	.66
	.6667	0.67
	0.666	
	0.667	.67



13



Mark for Review

A researcher surveyed undergraduate students, graduate students, and postdoctoral students. The number of undergraduate students surveyed was 7,150% of the number of postdoctoral students surveyed and the number of graduate students surveyed was 45% of the number of undergraduate students surveyed. If there were 6,435 graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?

Answer Preview:

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .66 .6667 0.666 0.667	0.66 .66 0.67 .67



14



Mark for Review

$$16x - 20y = 6y + 8$$

$$ty = 8x + \frac{1}{5}$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t ?

Answer Preview:



$$f(x) = 7(4)^x$$

The function f is defined by the given equation. $g(x) = f(x + 3)$, which of the following equations defines the function g ?

(A) $g(x) = 21(4)^x$

(B) $g(x) = 448(4)^x$

(C) $g(x) = 21(12)^x$

(D) $g(x) = 343(64)^x$



The bone mineral density of a certain person's thoracic spine was estimated to be 0.812 grams per square centimeter. What is this estimated bone mineral density in grams per square millimeter? (1 centimeter= 10 millimeters)

(A) 81.2

(B) 8.12

(C) 0.0812

(D) 0.00812



Which expression is a factor of $49p^{19} - 121p^{17}$?

(A) $7p + 11$

(B) $49p^2 + 121$

(C) $11p - 7$

(D) $-72p^{40}$

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5	
	3.50	3 1/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
	.6667	.66
		0.67



18



Mark for Review

$$\frac{5}{9}(5x + 9)(x + \sqrt{5k + 9})(x - \sqrt{5k + 9}) = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 72. What is the value of k ?

Answer Preview:



A right rectangular prism has a base area of $56t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{14}{3}\text{cm}$, and the height of the rectangular prism is 3 cm . Which expression represents the surface area, in cm^2 , of the right rectangular prism?

(A) $112t + 56$

(B) $184t + 28$

(C) $1,680t + 28$

(D) $168t$

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .66 .6667 0.67 0.666 .67 0.667	0.66 .66 0.67 .67



20



Mark for Review

In the xy -plane, a unit circle with center at the origin O contains point A with coordinates $(1, 0)$ and point B with coordinates $(\frac{5}{\sqrt{34}}, -\frac{3}{\sqrt{34}})$. If the measure of angle AOB is p radians, what is the value of $\frac{\cos p}{\sin p}$?

Answer Preview:

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
	.66	
	.6667	
	0.67	
	0.666	
	0.667	.67



21



Mark for Review

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{334})$. If $h(m) = 0$, what is the greatest possible value of m ?

Answer Preview:

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A computer repair specialist charges \$160 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 6 hours of repair is \$360. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

(A) $f(x) = 50x + 60$

(B) $f(x) = 50x + 160$

(C) $f(x) = 60x$

(D) $f(x) = 60x + 160$